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Privacy Preserving Cloth Try-On Using Mobile Augmented Reality

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Abstract—Virtual try-on applications make it possible for buyers to watch themselves wearing different garments without physically trying on them. The prevailing approach for virtual try-on has been based on virtual fitting rooms, in which several cameras are used to identify the skeleton and posture of a user in order to render a garment on the user's image. Although this approach has been implemented successfully using different techniques, the privacy of users can be compromised as some users might be reluctant to stand in front of cameras in a fitting room. This paper proposes an alternative approach that allows a customer to watch a 3D model of her/him wearing garments on a personal mobile device using Augmented Reality (AR). Among 3D human models that are automatically generated, a model selection technique is proposed that makes it possible to find the right size model representing the anthropometric features of the user. This approach is accompanied by body customization and face generation modules to generate a realistic representation. Several quantitative experiments as well as user studies were performed to evaluate the accuracy, efficiency, usefulness, and privacy of the proposed technique.

Index Terms—Virtual cloth try-on, augmented reality, body customization, privacy preserving cloth try-on.

1 Introduction

Virtual try-on systems make it possible for buyers to see themselves wearing different garments without actually wearing the clothes. Such a system can improve the shopping experience by assisting customers to make purchase decisions or at least help to narrow down selections before physical trying on them [1]. Various virtual try-on systems have been proposed in recent years (e.g., [1], [2], [3]). Generally, these techniques are implemented using Augmented Reality (AR), in which computer generated garments are superimposed on real data. In recent years, AR has gained significant attention in different emerging applications [5], [6]. As an interactive display technology, AR makes it possible to increase the understanding of viewed objects by combining real and synthesized information [4].

We classify virtual try-on systems into image-based, live AR, and model-based categories. The image-based technique is generally implemented in web-based solutions, in which the image of a user is augmented with the garment browsed at the website. In [1], segments from the picture of a model wearing a garment are shown on the picture of a user. In [7], a virtual try-on based on person's 2D images is proposed, in which an image warping technique is applied to map the apparel image onto the individualized body. Some techniques such as [12] reconstruct the garment from video sequences of the model wearing this cloth. Some methods such as [14] employ a technique to select the best matching pictures of poses in a previously recorded database that contains all poses. However, such techniques do not allow to see the textured images from arbitrary viewing angles. Digitizing the shape of a garment such that fits a user's body is a complex problem that may not be practical to be handled in real time. Drawing a textured clothes over the image of a user is one of the common techniques in AR try-on systems (e.g., [1]). The drawback of image-based virtual try-on techniques is the augmentation of pictures as static overlays, that does not allow generating arbitrary viewing angles. However, customers expect to see the garment from different viewing angles.

In live AR try-on, images captured by video cameras in a virtual fitting room are augmented with the images of garments in real-time. With the advent of motion capture tools (e.g., Microsoft Kinect), new opportunities have been created to increase the capabilities of virtual try-on systems. This varies from building 3D models by scanning the objects [8] to recognize human actions [9] using depth cameras. Producing 3D models of moving humans in real-time is a challenging task that requires multiple sensors. In [8], a technique is proposed to reconstruct the 3D model of a moving human using multiple-Kinect capturing system. However, mapping cloth textures on a moving model can still be a complicated task. This also requires a fitting room equipped with several Kinects that affects the convenience of users. In [10], a user's human model as well as clothes are constructed using multiple cameras, where the skeleton obtained from a Kinect is mapped to the user's movements. In [12], users standing inside a fitting room equipped with cameras can see themselves wearing different clothes on a large TV. In this technique, the system captures the user's image, while different garments are augmented on the image. The complexity of live AR try-on to process the images of users as well as difficulties to set up virtual fitting rooms are the main disadvantages of this method [1]. Although capturing the pose of a user provided by color-depth cameras such as Kinect may be sufficient in computer games and other applications, it is not able to accurately align virtual clothes with the user's body [12].

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In the model-based virtual try-on, garments are simulated on the reconstructed 3D model of a user. In [13], virtual 3D human characters wearing the target clothes are shown to the user. Alternatively, motion capture devices can be used to reconstruct the 3D models of clothes. In [13], a clothing design system is proposed that offers the assemble of a garment on the 3D model of a human model. Some methods such as [15] transfer garment meshes from 3D model of the model wearing the garment to the user's model. However, this is a complex process that needs shape and pose adaption, which may not be suitable for a mobile application.

This paper proposes a prototype for virtual try-on based on augmented reality (called PATRON), by which a user can watch a 3D model of her/him wearing previously recorded garments through scanning an AR-enabled tag by a mobile device. The user can turn around the tag or turn the tag to see his/her model from different viewing angles. To this end, a garments database storing 3D models of humans wearing garments as well as specifications of the users is automatically created. A model selection technique is proposed to select a 3D model matching the specifications of a user. To customize a body model based on anthropometric specifications of a user, a novel body customization technique is proposed to scale various parts of the model separately such that retains the harmony of the whole body. To generate a realistic model, a face generation technique is used to reconstruct the 3D face model of a user based on the input image and a reference face model.

Users are concerned about the data collected about their behavior (such as movements and poses) in front of a mirror. Unlike existing virtual try-on systems based on virtual fitting rooms that may compromise the privacy, neither the images nor the specifications of users are disclosed in PATRON due to the use of various privacy preserving policies. In summary, the contributions of this paper are:

- Developing a virtual try-on system based on AR that makes it possible to render a 3D model of a user wearing a garment on a mobile device.
- Introducing a privacy preserving protocol for virtual try-on systems by enforcing secret sharing and secure computation techniques.
- Proposing a novel body customization technique to generate 3D human models from a template model and anthropometric features.
- Conducting a user study to evaluate the user experience in virtual try-on systems.

2 PATRON

The privacy preserving virtual try-on system proposed in this paper is performed in offline and online phases. The overall architecture of the mobile AR for virtual try-on in these two phases are shown in Fig. 1.

2.1 Offline Phase: Model Database Construction

Generating a 3D human models, 2D images of models and 3D head models are performed in this phase.

2.1.1 Model Generation

Manually generating a collection of 3D human models wearing garments is impractical because of considerable time and expenses required for modeling. Moreover, models created manually are not generally accurate. To address this problem, we turn to the 3D scanning technique that provides a more accurate, economical and fast solution. Six common sizes in clothing including $\mathcal{Z} = \{XS, S, M, L, XL, XXL\}$) were taken into account. In the ideal case, in order to collect realistic models, the system needs to scan Z people of different sizes wearing G various garments. However, this requires scanning $M = Z \times G$ models, which may be not easy to extend the space of garments. To address this problem, only a medium size model-user is scanned. However, we use a 3D model generation technique through customizing the medium size model in order to generate the 3D models of other sizes.

In order to generate new body size models from a template (medium size model), the technique proposed in this paper employs the idea of local scaling (scaling of each part separately) and global scaling (scaling the whole body) proposed in [1]. However, the structure and the algorithm of the technique we propose for model generation is different. First, unlike [1] in which it is not clear how body segments are created, we use a body segmentation technique to automatically determine the main limbs of the body model. The main segments of the body model after the segmentation are left arm, left leg, torso, right arm and right leg. These segments make it possible for independent local scaling of the parts. Second, unlike [1], we formally show how scaling functions affect the vertexes and meshes in the model. In particular, we generalize different scaling functions as a scaling deformation function that makes it possible to define local scaling functions. Third, unlike [1] that employs the height of a user for global scaling, we define a new global scaling function based on a harmonized mean of weight w and height h as two parameters defining the shape of a body.

Scanning Model-Users. The medium size model-user wearing different garments was scanned by a 3D scanner using 3DBuilder and Kinect. Due to the limited resources were available for setting up the experiments, the virtual try-on we propose is for male garments. The model-user was scanned in the standing position, hands down and watching front. As shown in Fig. 2, the RGB as well as the depth cameras in the scanner make it possible to generate 3D models in which the textures of garments can also be rendered. The model-user was asked to wear 24 different garments including 6 shirts, 6 sweaters, 6 jacket and 6 jeans.

Deformation Function. In order to generate the 3D models of sizes other than the medium size, additional resizing information is required to deform the mesh. We extracted this data from 48 participants (the average of 6 participants for each size). Given the medium size model as template, we argue consistent combination of deformations on several segments makes it possible to generate human models with different body size. Changing the length of leg, arm, and torso requires scaling operations. Formally, let M the segment of a body mesh extracted using the body model segmentation technique. Given the vertexes of this segment ($V = \{v_i\}_{i=0}^{n} \subset R^3$), each point v_i is defined as

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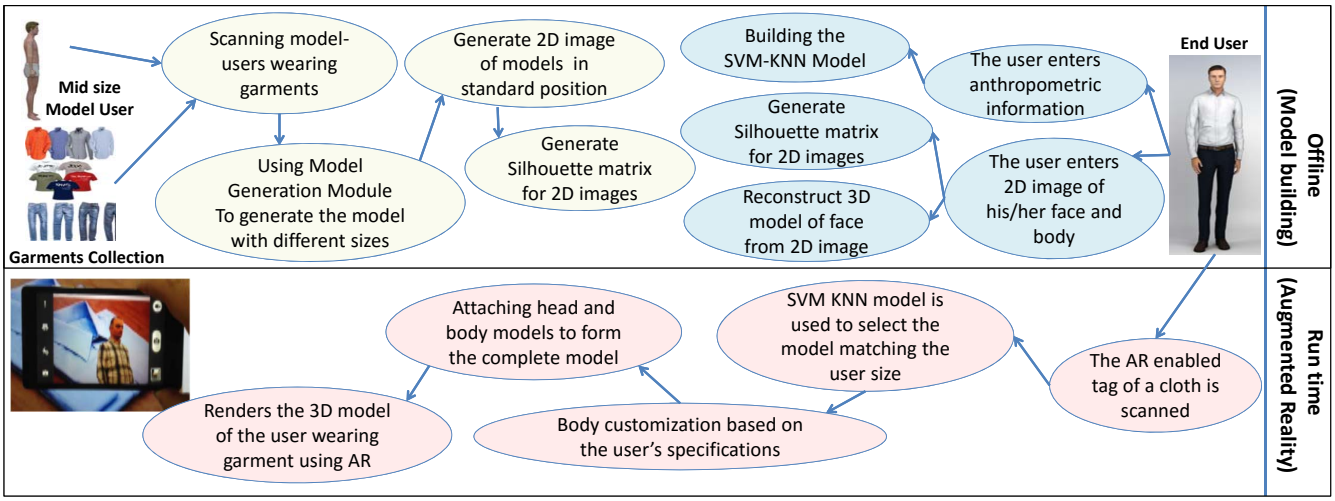


Fig. 1. The overall architecture of the privacy preserving virtual try-on system (PATRON) including offline as well as runtime activities.

$v_i = (x_i, y_i, z_i)$, where n is the number of vertexes in the segment. We identify the scaling deformation function as:

$$d(v_i) = d_i = v_i + \sum_{j=1}^{k} s_j f_i()$$

where d_i is the new position of v_i after deforming the segment M, and k is the number of deformation functions applied on this segment. The scaling is achieved through parameter s_j, which is the scale factor of the jth deformation function (note that s_j is triple $\langle s_x, s_y, s_z \rangle$ that defines the scale factor on different dimensions). The model customization is realized using the deformation function applied on the template model (medium size model) based on the anthropometric properties. Formally, the deformation procedure is defined as:

1. Each vertex M_i of M is parameterized on triangle b_j=$\triangle(p_{j0}, p_{j1}, p_{j2})$ using local coordinates (u_{ij}, v_{ij}, w_{ij}) such that:

$$M_i = p_{t0} + u_{ij}.U_j + v_{ij}.V_j + w_{ij}.W_{ij}$$

2. After deformation, triangle b_j is changed to f_j=$\triangle(q_{j0}, q_{j1}, q_{j2})$ such that vertex M_i is changed to $m_{ij}^{'}$:

$$M_i^{'} = q_{t0} + u_{ij}.U_j + v_{ij}.V_j + w_{ij}.W_{ij}$$

Body Customization. Customizing 3D avatars is a challenging task that can be achieved by realistically modifying a generic model based on some parameters, is an economical and effective way in comparison to creating models with different body shapes using the scan of a large number of model-users. Similar to [1], we employed twelve key human body measurements, including height, shoulder width, bust girth, waist girth, hip girth, thigh girth, ankle girth, waist height, crotch height, knee height, upper arm length and forearm length for customization. These key anthropometric features provide compact parameters for body geometry representation. The body customization is performed in the 5 following steps.

step 1: Considering weight w and height h as two main parameters defining the shape of body, the model is scaled globally. In order to apply w and h simultaneously, the harmonized mean of w and h is computed as:

$$s = 2\frac{w/w' \times h/h'}{w/w' + h/h'} \quad (1)$$

Then, the global scaling function $ScaleGlob(\mu.s)$ is used on the model, where μ is the normalization parameter computed heuristically based on display settings.

step 2: Using piecewise cubic polynomial interpolation, the legs are scaled along the y-axis based on the users waist w and crotch c. In particular, the function $ScaleLegs(\alpha.w.c)$ is applied on the legs, where α is the piecewise cubic polynomial parameter with the boundaries of 0 and h_{bust} (h_{bust} is the height of model's bust).

step 3: The torso is scaled along the y-axis but based on the user's waist w and knee heights k $ScaleTorso(\beta.w.k)$, where β is the is the piecewise cubic polynomial parameter.

step 4: The torso and the legs are scaled using the shoulder width, bust, waist, hip, thigh and ankle girths. First, arms vertices are translated along the x-axis, based on shoulder width. Then, the torso arms and the legs are scaled in the x-z plane. In this model, the girth change for the torso is considered as two internally tangent circles, where the scale is computed as the deviation from the uniform change from the outer circle to the inner circle such that:

$$k = (s_g - t)sin(\theta) + \sqrt{(s_g - t)^2 sin^2(\theta) + 2s_g t - t^2} \quad (2)$$

where t is a piecewise cubic polynomial function of y ($min(y) = 0, max(y) = neckHeight$) , and s_g is the ratio of the user's girth to the models one.

step 5: The arms are scaled based on harmonized mean of user's upper arm length u and the forearm length f using $ScaleArms(\lambda.u.f)$, where λ is the is the piecewise cubic polynomial parameter.

Using the body customization technique, we formed a repository of 3D human models $\mathcal{M}$ including 144 body models (24 different garments in six different sizes). Given a model $M_j^i \in \mathcal{M}$, the index j represents the identification of the garment, and i is the index of size.

2.1.2 Generate 2D Images

In order to extract the best 3D model matching the size of a user, a mechanism is required to compare the 2D image

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Fig. 2. Three scans of a model-user wearing different garments.

of the user with 3D models in $\mathcal{M}$. One possible solution is to directly map 3D models and images and compare the quality of mappings as a measure of similarity. However, this is a complex process that may not be appropriate for mobile platforms. In order to compare a 2D image and 3D model, we propose and use a technique in which a 2D image is generated from a 3D model. This way, the problem turns to the problem of comparison between two images.

Two images of human bodies can be compared in terms of size, if they are taken from the same view. To this end, images are captured from the front view of 3D models. These images are compared with the image of the whole body of a user from the front view (similar to the view of capturing images from 3D models). Before comparison, all images are cropped automatically such that no margin is left from the sides of images. In particular, given a model $M_j^i \in \mathcal{M}$, we first generate a 2D image img_j^i with the height of h and the width of w. Given an image img, we generate a binary image (pixels are either black or white) represented through the binary matrix I such that, if a pixel of the image is part of the body, then its corresponding item in the matrix is 1; Otherwise it is 0. The matrix I is defined as:

$$I[i][j] = \begin{cases} 1, & if\ img[i][j]\ represents\ a\ pixel\ of\ body \\ 0, & if\ img[i][j]\ represents\ a\ pixel\ of\ background \end{cases}$$

where $0 \leq i < w$ (width of the image) and $0 \leq j < h$ (height of the image). Generating the 2D binary silhouette matrix tends to be sensitive to the background texture or clutter especially for the case of mobile device applications. In addition, the assumption of white background may not be always satisfied in practice. To address this problem, we employ a background removal algorithm before generating the silhouette matrix of an image. In particular, we employed an image background removal library of OpenCV[1], which is one of the promising off-the-shelf libraries for this purpose. In this technique, after removing the shadows, the image is converted to a gray-scale image. Then, Gaussian blur and Canny edge detector techniques are used to detect the main area. The acceptable performance of this technique was the main reason to employ this library in PATRON.

2.1.3 Generate the Silhouette Matrix

In order to simplify the comparison between two images of human bodies (for the sake of estimating the best matching size), instead of comparing the images, we from and use *silhouette* images. More specifically, given matrix I representing the image matrix of an image img, we form the silhouette matrix S , such that approximates the shape of the human body. To this end, the image matrix is partitioned to n rows and m columns. As shown in Fig. 3, the image is partitioned to $m \times n$ rectangles with the width of w/m and height of h/n pixels (in our experiments m and n are heuristically considered 20 and 50, respectively). To form the silhouette matrix, an average of the items in each rectangle s_{ij} is computed as:

$$s_{ij} = \sum_{i=0}^{[w/m]} \sum_{j=0}^{h/n} I_j^i / (w/m * h/n)$$

Then, the silhouette matrix S is formed from the average score of the rectangles such that:

$$S[i][j] = \begin{cases} 1, & if\ s_{ij} \geq 0.5 \\ 0, & if\ s_{ij} < 0.5 \end{cases}$$

The silhouette matrix is used in the model selection module at run time, where we propose a technique to find the best 2D image of 3D models matching the size of a user.

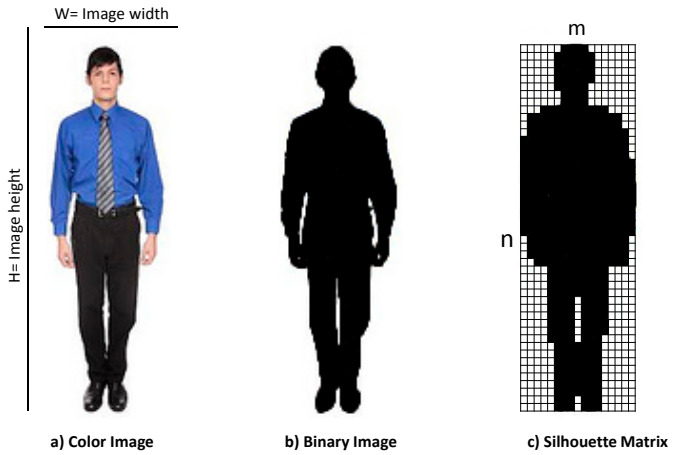


Fig. 3. color image(a), brainy image(b), silhouette image of a model-user

2.1.4 Reconstructing the 3D Model of Head

The model of a user will not be complete without the 3D model of the face. One option would be using a 3D scanner to generate face models. However, this may not be convenient for users of our application as this requires extra equipment and predefined settings to scan the head of users. One alternative solution would be using a human model without a head, or simply using a general avatar. To seek for a technique that takes into account the convenience of users in generating 3D models while creating accurate models, we adapted the face reconstruction technique proposed in [17]. Modifications were made to make it possible to generate the model in a format that can be rendered in this platform. In this technique, a 3D face model is reconstructed from a single image and a reference face model. A input face image is used to mold the reference 3D model in order to reconstruct the model that simulates the face. Shading information with prior knowledge of the single reference model is combined to recover the 3D shape of the face in the image.

1. http://www.docs.opencv.org

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2.2 Runtime: Model Selection and AR Rendering

PATRON is activated by scanning the tag of a garment in retail stores or online. Once the tag (i.e., AR marker) is identified by the application, the system needs to find the model $M_j^i \in \mathcal{M}$, such that represents the garment j with size i and simulates the body of the user in the best possible way. Then, this model is customized to represents the actual size of the end user. Finally, the model of head and body are attached and smoothed to form the final model. In the proposed application, colors are selected by following the guidance of the opponent process theory of color [18].

2.2.1 Model Selection

Selecting a model matching the size of a user among the models generated by the model generation module ensures the minimum distortion when the model is modified based on the specifications of the user. In order to select the appropriate model, given the silhouette matrix S_a (corresponding to the 2D image of a user), we need to find the silhouette matrix S_b from the repository of all silhouette matrices (corresponding to the 2D images of the 3D models in $\mathcal{M}$) such that the difference between S_a and S_b is minimum. This turns the problem into an image classification.

Different techniques can be exploited to tackle the image classification problem. Among which, Support Vector Machines (SVM) have gained increasing popularity because of the high generalization capability [19]. This technique has been empowered using different feature descriptors. Formally, the image classification problem can be defined as follows. Given a set of traing data L = $\{x_i, y_i\}_n^{i=1} \in X^n \times Y^n$ of n images along with its labels $y \in Y = \{1, \ldots, c\}$, the SVM classifier is defined as:

$$y^\star = \underset{k=1,\ldots,c}{\operatorname{argmax}}\, g_k(x) = \sum_{i=1}^{n} \alpha_i^k\, (x, x_i) + b^k$$

where c is the number of classes, and $y^\star$ is the class label suggested by the SVM classifier. In this formula, $g(x)$ is the discriminant function and k is the index of this function, and the set $\{\alpha_i^k\}_n^{i=1}$ are signed dual coefficients.

The main drawback of this technique is a large number of computations during testing due to the need for matching each test image against a large number of images. In particular, to classify an image using SVM, the kernel values between the test image and Support Vectors (SVs) must be computed. Our technique requires a SVM technique that is fast enough to run on mobile devices (with resource limitations), while preserving the accuracy of classification. In this paper, we employ the SVM-KNN technique proposed in [19], which is a hybrid of K-Nearest Neighborhood and SVM. The experiments showed the accuracy of this technique with a reasonable computational cost that is suitable for our technique. The main idea behind this technique is the narrowing of training data using KNN before applying SVM. In this technique, the kernel is:

$$K(x, y) = (\langle x, x\rangle + \langle y, y\rangle - \langle x - y, x - y\rangle)/2$$

where the distance functions d of the KNN classifier is transformed to the kernel of SVM such that:

$$K(x, y) = (d(x, 0) + d(y, 0) - d(x, y))/2$$

Given a test image as query, a collection of K_{s1} neighbors based on the distance function is extracted. Then, the pairwise distance of the union of K neighbors and the query are computed. This pairwise distance matrix is converted into a kernel matrix employing the kernel trick. Finally, DAGSVM algorithm (a high-speed algorithm to implement multiclass SVM) is applied on the kernel matrix and the appropriate label of the test image is assigned. Originally, SVM-KNN takes into account shape and texture as two major cues of images for classification. However, since we classify silhouettes, we only apply shape measures as there is no need to apply the texture measures. In particular, we use Shape context based distance and Geometric blur based distance for this purpose. In the evaluation section, we show how SVM-KNN outperforms naive *direct silhouette matching* and *feature-based model selection* for image classification.

2.2.2 Final Body Customization

As discussed in [10], virtual try-on applications must be able to deliver great realism in the visual quality and the interaction techniques in order to be accepted by users. Generating a 3D model representing the body of a user in the best possible way is crucial to achieve this goal. Although in the model selection step the 3D model of a user matching the size of the user is selected, these models must be modified at runtime to match the body of a user. Anthropometric features can be used to modify the body models with a satisfactory precision. Given the garment g, the model selection step makes it possible to find the 3D model of the model-user wearing the garment with the size s_i matching the size of the user. In the final body customization step, the selected model is modified based on the specific anthropometric features of the user in order to achieve a model that represents the user in the best possible way. This is an optional step that allows to improve the model shown to the user at the cost of extra processing time.

2.2.3 Attachment and Smoothing

The final step in making user's model is attaching body (the output of body customization) and head (generated in Section 2.1.4) models. The model attachment must be performed in a smooth way such that the final model looks like a whole body. To achieve this goal, two issues must be taken into account. First, the neck is scaled along the x-axis and y-axis based on the harmonized mean of the neck size of the body and the neck size of the head. This results in matching the connection plane between the models of body and head at x-z plane. The output of this step is the 3D model of the user wearing the garment.

Local deforming a body segment can result in discontinuous passes at the boundaries of two regions. In order to address such irregularities, boundary vertexes must be deformed smoothly. We applied the Tukey window [11] as a smoothing function. This function is in charge of reducing the displacement value on the boundaries of two segments. Given $W(u)$ the matching value on y axis and N is the number of samples on x axis, $w(u) = 1$ when $0 < |u| < (\alpha N/2)$ and $w(u) = .5[1 + cos(A)]$, when $(\alpha N/2 < |u| < N$. In this formula, $A = \pi u - (\alpha N/2)/(2(1 - \alpha)N/2)$, and $u = L(V)_t N$. $L(v)$ is the normalized local coordinate function such that $L : R3 \rightarrow R3$, and $0 \leq L(v)i \leq 1$ for $i = 1, 2, 3$. The amount of smoothness is indicated by the value of α that determines the ratio of taper on constant sections.

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Fig. 4. The augmented reality view of the PATRON. By scanning the AR-enabled tag on the cloth, the 3D model of the user wearing that cloth is shown on the screen. The user can see the model from arbitrary viewing angles by turning the device around the target or moving the target.

2.2.4 Rendering the 3D Model Using AR

PATRON is implemented using Augmented Reality (AR). More specifically, we use a marker-based AR, in which, vision-based tracking techniques relying on the natural characteristics or visual cues of the environment are used to perform tracking. In this technique, tracking is performed through fiducial marker detection, feature detection or edge detection. To this end, the current camera pose must be computed to indicate how to augment real image frame with graphical objects. PATRON is implemented using Processing and the open source library of NyAR4psg.

3 Privacy Preservation

According to [10], privacy is one of the important reasons that buyers prefer online buying and selling clothes. Personalization, which is known as tailoring a service or a product to accommodate particular needs of individuals has a conflict with privacy. Generally, personalization cannot be achieved without some loss of privacy of consumers. This is because personalization needs information of users to tailor offerings to consumer's tastes. In terms of virtual try-on, publicly disclosing the image of a user, anthropometric features, or 3D model of the user wearing garments are privacy concerns that must be addressed. In this paper, two main privacy preserving policies are enforced to make sure preserving the privacy of users in PATRON. First, we use a secret sharing technique to split the private data using randomized functions. This way, a part of sensitive data is kept on the mobile device, while the other part is stored on the server. Using this technique, it is not possible for either server or client to gain the real value without the value of other part. Second, we use a private secure computing technique to make sure the computations of the model modification functions are performed securely on the server. The technique we propose is used in a semi-honest setting indicating that server and client do not deviate from the protocol specification. However, they may try to gather information out of the protocol.

3.1 Secret Sharing between Mobile Device and Server

Let $X = x_i, , x_n$ representing n piece of information entered by the user among registration regarding the anthropometric measures. For each $x_i \in X$, we generate random r_i such that $-x_i \leq r_i \leq x_i$. Using this random number, x_i is spitted to two parts including p^s_{xi}, which is hold on the server, and p^c_{xi}, which is hold one the client, such that $p^s_{xi} = r_i$ and $p^c_{xi} = r_i - p^s_{xi}$. Using this technique, a secret sensitive is distributed amongst server and the mobile device (can be reconstructed only when a party have both parts).

The facial image of a user is also a sensitive information that must be protected from malicious accesses. To achieve this goal, we use the secret image sharing technique proposed in [16]. In the general case of this technique, the secret image is used to generate n shadow images, where any r or more shadow images are enough to regenerate the secret image. In our case, we generate two shadow images ($r = 2$), where both of them are required to reconstruct the secret image. The client is in charge of generating the shadow images r_1 and r_2, where r_1 is stored on the client and r_2 is kept on the server. In the case of face generation, the face generator component reconstructs the facial image using r_2 shadow image on the server as well as the r_1 shadow image received from the client. The original reconstructed image is deleted after time t in order to make sure exploiting the already generated image before t.

3.2 Secure Computation

In PATRON, the client C (mobile device) asks the server S to evaluate a function f over C's private input x. Function f can be any deformation function (e.g., scaleTorso), and $x = (x_1, ..., x_n)$ of different values $\{x_i\}^n_{i=1}$. The set of anthropometric measures of C are considered sensitive in our architecture. Through the secure client-server computation, S is in charge of computing f, while x and $f(x)$ remain private. x_is are encrypted under C's public key p, where the corresponding cipher-text of x_i is c_i. The final result of encryption function $ENC_p(f(x_s, x_1, x_2, \ldots, x_n))$ is c, which is computed via the evaluation algorithm $Eval$. In this formula, x_s is the private value supplied by S. The parameter α in $ScaleArm$ and β in $ScaleTroso$ are examples of the the parameters provided by S.

The evaluation algorithm $Eval$ is associated with key generation, encryption, and decryption. In order to implement the $Eval$ algorithm, we employed the Fully Homomorphic Encryption (FHE) scheme [20]. In an FHE scheme, any function can be executed without revealing the input information or outputs. Generally, FHE algorithms are used to remotely perform the computations on semi-honest servers on the cloud.

4 Evaluation

To evaluate PATRON, we first measure the accuracy of the model selection component. We also measure the efficiency

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of our techniques in terms of execution time. A user study is also performed to evaluate the effect of using body customization on the performance and the experience of users in virtual try-on. We also examine the consumer adoption of our virtual try-on technology. We empirically measure both quantitative and qualitative data related to these systems in order to make well supported statements regarding the effect of model selection and model customization techniques for AR mobile applications in virtual try-on. We compare our model selection technique with two baseline techniques including *direct silhouette matching* and *feature-based model selection* described in the following.

Direct Silhouette Matching. In this technique, given the silhouette matrix of a user's image, the silhouette matrix of the 2D image of the 3D model with the minimum difference is selected. Formally, given the silhouette matrix S, the silhouette matrix S_s^g corresponding to 3D model $M_s^g \in \mathcal{M}$ (the 3D model of the model-user with size s wearing the garment g) is selected such that:

$$S_g^s = \text{argmax}_{M_j^i \in \mathcal{M}} \sum_{i=1}^{m} \sum_{j=1}^{n} S_i^j \oplus S$$

Feature-based Model Selection. In this technique, anthropometric features of a user are used to find the best model matching the size of a user. In this probabilistic inference model, given a known set of features $(f_1, f_2, ..., f_k)$ that are associated with the size of models, it is possible to estimate the prior probabilities for each feature expressing the size of a user. Using Bayesian inference makes it possible to compute the posterior probability of the size of a user given the features of this user. In this model, s is a categorical class variable whose domain is $\mathcal{Z} = \{XS, S, M, L, XL, XXL\}$, and the features $f_1, ..., f_k$ are the evidence variables, representing properties of models. With the assumption that evidence variables $\{f_1, ..., f_n\}$ are conditionally independent (that may not be valid assumption), the model is:

$$Pr(s|f_1, ..., f_k) = \frac{Pr(s) \prod_{i=1}^{k} Pr(f_i|s)}{Pr(f_1, ..., f_k)}$$

where the size maximizing $Pr(s|f_1, ..., f_k)$ is the most probable size given the specifications of the user ($s = \text{argmax}_{s_i \in \mathcal{S}} Pr(s_i|f_1, ..., f_k)$).

4.1 Accuracy

We used the information of 60 male participants to evaluate the accuracy of our model selection technique. We captured an image of the standard pose of each participant as described in Section 3.1.2. These images were fed into the SVM-KNN and direct silhouette matching model selector modules. In the case of feature-based selection, the Bayesian model was trained with anthropometric features of users measured manually. We measured the difference between the actual size of a user and the size selected by the image-based model selector as a measure to evaluate the accuracy. More specifically, given the sequence of the sizes $\mathcal{Z} = (XS, S, M, L, XL, XXL)$, the difference between two sizes $s_i, s_j \in \mathcal{Z}$ is $e_{i,j} = |position(s_i) - position(s_j)|$, where $position(s)$ is the index of s in $\mathcal{Z}$.

The distribution of errors (the differences between the actual size and the selected size) is shown in Fig. 5. According to the results, our SVM-KNN model selection technique

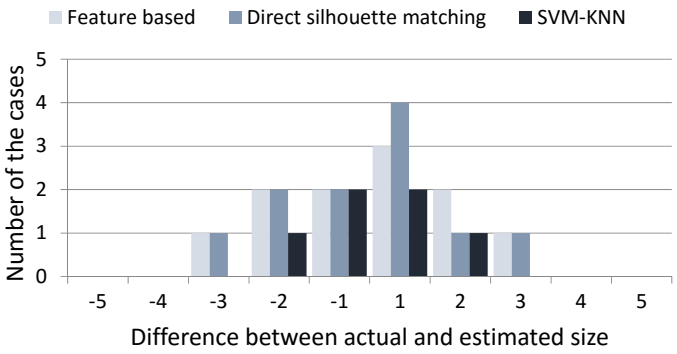


Fig. 5. The histogram of errors in different model selection techniques.

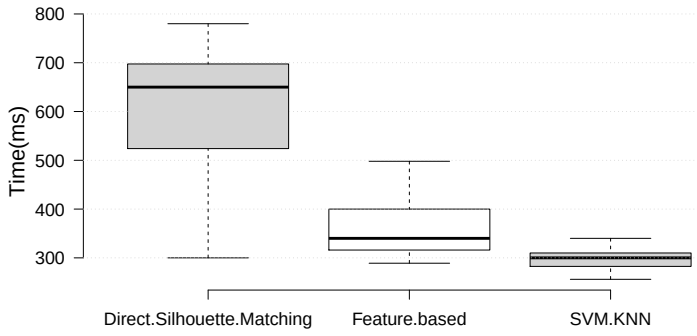


Fig. 6. The execution time of model selection in different techniques.

outperforms the feature-based as well as direct model selection techniques. More specifically, the SVM-KNN model selector achieved the accuracy of 91% where this value is 87% for direct silhouette matching and 76% for the feature-based model selection. We argue although the results for the feature-based selection can be improved by increasing the number of training data, in practice and in real scenarios, inaccurate measuring the anthropometric features by users can still result in impairing the final result.

4.2 Performance

Since PATRON is a mobile application, execution time and power consumption are big challenging issues that must be addressed. To this end, we evaluate the computational efficiency of PATRON and compare the results with the efficiency of direct silhouette matching and feature-based model selection techniques. PATRON is a dynamic application, where body models are built on the server and transferred to the client once an AR-enabled tag is scanned. Consequently, model selection and body customization steps are performed on the server. In order to directly compare different model selection techniques, what we report here as an execution time is absolute time for model selection. As shown in Fig. 6, the execution time of SVM-KNN model selection is almost the same as the execution time of the feature based model, while it is at nearly a quarter of the execution time of direct silhouette matching. The results correspond to running PATRON on a Huawei Honor 3 with android 4.2.

Run-time execution time includes the time required to scan the augmented reality enabled tag, connection time,

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data transformation time, reconstructing the 3D face model, attaching the head and the body of models, smoothing the model, and rendering the model. An average of execution times for 50 runs are shown in Fig. 7. Scanning a user-model which is performed offline takes around 10 seconds. Note that the final body customization is an optional module that is performed upon the request of a user to generate a 3D model matching the specifications of the user. Due to the computations conducted at runtime to locally and globally scale the mode, this is achieved at an extra cost of processing, which is significant in comparison to the total model selection process.

4.3 User Study

We present and compare three different scenarios that are combination of Model Selection (MS), Body Customization (BC) and Face Generation (FG). These scenarios are:

- sc_1 (+MS-BC-FG): model selection without body customization and Face generation.
- sc_2 (+MS+BC-FG): model selection with body customization but without face generation.
- sc_4 (+MS+BC+FG): model selection with body customization and face generation.

We conducted a user study to evaluate the effectiveness of these scenarios and the perception of quality attributes in end users. These measures cover mental processes when using the system, ease of use, and perceived usefulness of the system. We used an unsupervised approach to conduct the study, where users were free to browse the clothes and virtually try-on any cloth they want. The study was started after gaining an informed consent of participation in the study. A pre-study questionnaire was filled by each participant to gain insight about educational level, prior experience with augmented reality applications and virtual try-on systems.

Measures. We designed and performed a within-subjects user study to determine the advantages and disadvantages of different scenarios. We considered the type of system (i.e., sc_1, sc_2, and sc_3) as an independent variable. To measure and compare perceptual and usability issues between different scenarios, we employed 15 attributes proposed in [1], which are categorized in three groups including Quality Attributes (QA), Cognitive Attributes (CA) and Attitude Towards Using (ATU). The purpose of the quality attributes is to evaluate to what extent the application can assist a purchase decision. On the other hand, the purpose of cognitive attributes is to measure mental process involved when using the system (e.g., learning attention and emotive elements). Attitude towards using covers a set of attributes related to the usefulness and the ease of use.

Participants. A total of 60 individuals (those were used to evaluate the accuracy of PATRON) with an average age of 25.2 were recruited to participate in this study. Having an experience of working with smartphones was the precondition to participate in the study. According to the pre-study questionnaire, none of the participants had used virtual try-on applications before. Over 95% of them reported that they have had experience of online shopping.

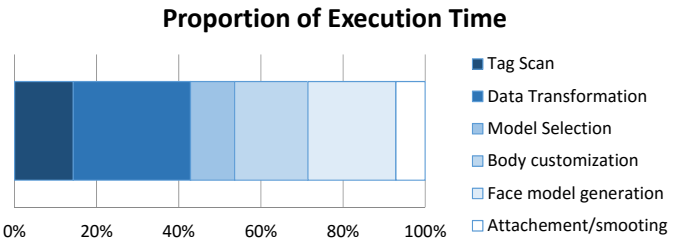


Fig. 7. The execution time of different procedures in PATRON

Results. In order to address the potential learning effect, the order of exposure to the different scenarios was varied, resulting in the assignment of participants to three different groups. In particular, we formed 6 groups of participants (10 participants in each group), where the members of different groups were exposed to a different order of scenarios. This way, we removed the order in which users work with different systems as an independent variable. Participants were asked to virtually try-on any different clothes among 24 clothes for which 3D models were available. The garments were put on a table in a room with controlled light and temperature. They answered the statements regarding QA, QC, and ATU using a 5-point Likert scale, to describe to what extent they are agreed with the statements.

Table1: Pair-wise comparison of the measure of the quality attributes using Wilcoxon-Mann-Whitney tests

Quality Attributes (QA)	Wilcoxon-Mann-Whitney
sc_1, sc_2	Z= -2.43, p = 0.00453
sc_1, sc_3	Z= -1.12, p = 0.00847
sc_2, sc_3	Z= -0.21, p = 0.85452
Cognitive Attributes (CA)	Wilcoxon-Mann-Whitney
sc_1, sc_2	Z= 0.85, p = 0.356452
sc_1, sc_3	Z= 0.28, p = 0.656425
sc_2, sc_3	Z= -0.55, p = 0.956456
Attitude Towards Using (ATU)	Wilcoxon-Mann-Whitney
sc_1, sc_2	Z= -3.43, p = 0.00014
sc_1, sc_3	Z= -2.34, p = 0.00271
sc_2, sc_3	Z= -0.35, p = 0.79568

Pair-wise Wilcoxon-Mann-Whitney tests were conducted to identify statistically significant differences between different scenarios (See Table 1). The Z score is a measure of standard deviation that is used to indicate the null hypothesis must be rejected. The p-value is the probability that the null hypothesis is falsely rejected. Based on the confidence level of 95% (confidence level = 0.05), the critical Z score values are -1.96 and +1.96 standard deviations. According to the $Z = -2.43$ for difference between sc_1 and sc_2 in QA, the Z value is less than -1.96 and p is less than 0.05. Consequently, there are statistically significant differences between sc_1 and sc_2 in QA. Accordingly, there are statistically significant differences between sc_1 and sc_3 in QA. The results show the importance of body customization to assist purchase decisions in spite of extra processing time required to customize the 3D models based on anthropometric measures. However, no statically significant differences were found between sc_2 and sc_3. We argue this is due to the low quality face models generated.

For CA, there were no statistically significant differences

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between scenarios. For, ATU, there were statistically significant differences between sc_1 and sc_2, and also between sc_1 and sc_3. This shows the importance of body customization in user's attitude towards using (related to the usefulness and the ease of use). In addition, the results state a stronger affinity of participants towards customized models, where they valued the models that were close to their size.

4.4 Privacy

We conducted a user study to measure the perception of users from privacy and the effect of this issue on using virtual try-on systems. We designed a questionnaire including 8 randomly ordered statements using 5-point Likert scale (1= strongly disagree and 5= strongly agree) to describe how important privacy is for the users of virtual try-on systems. The questions are about sharing and using images, 3D models, and anthropometric features of users. Through this questionnaire, we aim to find if privacy is an important factor in the acceptance of the virtual try-on technology. The overall mean of 1.53 for our 60 participants represents the importance of privacy concerns for users employing virtual try-on systems. We were also interested to find if the general level privacy for online behaviors is correlated with their attitude towards the privacy of virtual try-on systems. To this end, we used the questionnaire proposed in [21] to measure general privacy concerns of users in online behaviors. This questionnaire includes six questions about the general privacy caution and six questions about the technical privacy protection. According to the Pearson Correlation test between general privacy concerns and the expected privacy for virtual try-on system (correlation = -0.59, p = 0.005), the considerable negative correlation shows that users who are cautious about privacy issues are also careful to avoid potential privacy problems.

5 Conclusion

In this paper, we proposed a privacy preserving virtual try-on based on augmented reality (PATRON). In this system, once a user scans the tag of a garment using her/his mobile device, a 3D model of the user wearing that garment is shown on the cloth tag using augmented reality. This way, a user can see the model from arbitrary viewing angles either by turning the camera around the tag or turning the tag itself. Secret sharing and secure computation policies are enforced in model selection and model customization procedures to ensure preserving the privacy of users in terms of personal photos, models and anthropometric information.

The experiments showed the high accuracy and efficiency of this technique in comparison to direct silhouette matching and feature-based model selection techniques. A user study to evaluate the effectiveness of body customization and face generation modules showed the importance of using body customization to provide realistic models of users. On the other hand, users did not value the face reconstruction module as expected. We argue this is due to the longer processing time and deformations of the face models, which is not able to generate a perfect model of a face from a single image. For future work, we aim to support simultaneously trying two or more garments (e.g., trying on a shirt and a jean at the same time).

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